Next, Griffith heat-killed the S strain bacteria and injected into mice, but the mice stayed alive, which suggests that heat killed S- strain are avirulent (inefficient to cause the disease). Then, he mixed the heat-killed S and live R strains and injected into the mice. The mice died. In addition, he found living S strain bacteria in dead mice (Fig. 7.9). This suggests that some transforming principle is present in the S form that has mediated the transformation of R form to S form. Later, the experiments conducted by Avery, Macleod and Mc Carty in1944 showed that the transforming principle is DNA, which carries the genetic information from one generation to another.

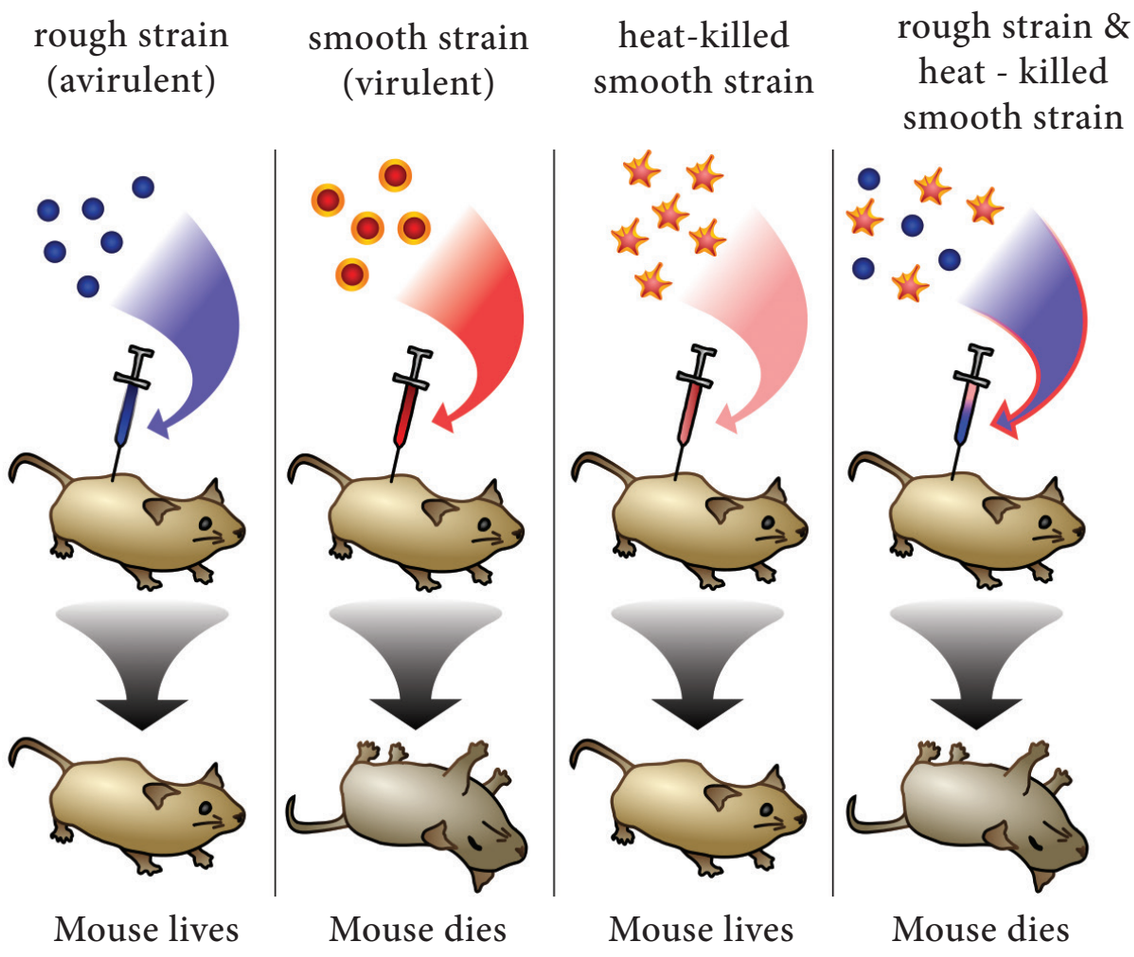


Figure. 7.9 Griffith's Experiment to identify genetic material

Functions of DNA

1. DNA is the genetic material of all living organisms except for RNA viruses like HIV. It is the greatest super chip and enormous data can be stored in high density.
2. DNA inside the zygote contains all the information required for developing into an individual organism.
3. DNA inherited from their parents is responsible for the characteristic features of an individual, i.e. shape of the eyes, ears, nose, colour of the skin, height, longevity, ability

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